

Storyboard: Bridging the Gap: From Innovation to Patient Impact

Ethics, Evidence, Regulation and Responsible Medical Device Translation • Dr Vikas Kumar Jha • KIIT University, Bhubaneswar • 35–40 minutes (≈35 min scripted + 3–5 min flexible discussion); scripted total 35:00

Story spine. We follow two composite patients, Ramesh (Parkinson's, freezing of gait, FoGO) and Kamala (knee osteoarthritis, SwaKnee), along a six-stage path: Need → Evidence → People → Permission → Access → Safety. Each stage answers one patient question. Every case claim is tagged Demonstrated, Reported (preliminary), Planned or Not established.

Teaching lenses. Stanford Biodesign (need-led process, Principled Decision-Making) • Harvard bioethics (HMS Center for Bioethics, MRCT Center, Petrie-Flom) • Cambridge (Engineering Design Centre inclusive design, Engineering Better Care systems approach) → adapted to India (CDSCO MDR-2017, ICMR 2017, MvPI, NPPA, UCMPMD 2024, DPDP).

Evidence key Demonstrated Reported, preliminary Planned Not established



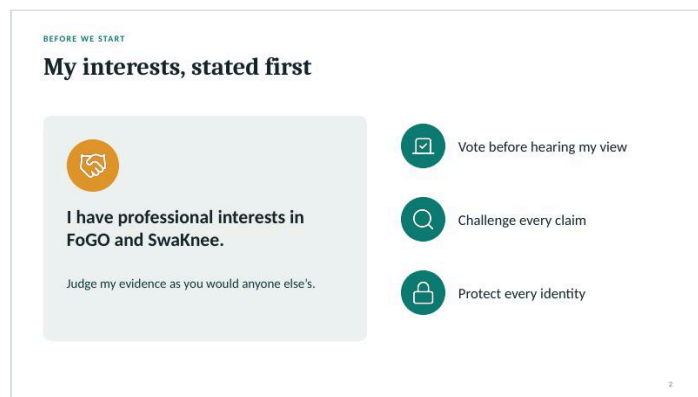
SLIDE 1 • OPENING • 0:00–0:45 • 0.75 MIN

Bridging the Gap: From Innovation to Patient Impact

Visual Dark title; authentic photographs of the FoGO ankle module and SwaKnee controller with applicator

Interaction Opening question to the room

Sources Supplied FoGO project material; SwaKnee product photographs



SLIDE 2 • OPENING • 0:45–1:15 • 0.5 MIN

My interests, stated first

Visual Disclosure panel + three ground rules with icons

Interaction Sets norms: vote first, challenge claims, protect identities

Sources Stanford GSB case on physician COI (2011); HMS COI policy

OPENING VOTE

Would you let a patient use it?

 Working prototype

 Promising lab results

 Test licence granted

1 Routine care

2 Research study

3 Not yet

Show 1, 2 or 3 fingers

SLIDE 3 • OPENING • 1:15–3:15 • 2.0 MIN

Would you let a patient use it? (opening vote)

Visual Three facts vs three answer cards (1 Routine care, 2 Research study, 3 Not yet)

Interaction OPENING VOTE by fingers; record split on a flipchart

Sources MDR-2017 (test licence ≠ permission to treat)

THE LENSES WE BORROW

Three teaching traditions, one patient in India

**Stanford Biodesign**
Start from a validated need

**Harvard bioethics**
Respect persons and their choices

**Cambridge design**
Design for people and whole systems

**Adapted for India** CDSCO • ICMR • MvPI • NPPA • out-of-pocket care

SLIDE 4 • OPENING • 3:15–4:00 • 0.75 MIN

Three teaching traditions, one patient in India

Visual Three lens cards converge on an India banner (CDSCO • ICMR • MvPI • NPPA • out-of-pocket care)

Interaction —

Sources Stanford Biodesign process and Need Statements brief; HMS Center for Bioethics, MRCT Center, Petrie-Flom; Cambridge EDC Inclusive Design Toolkit, Engineering Better Care (Clarkson), CCHLE

OUR PATIENTS TODAY

Two people we will follow

**Ramesh, 71**

 Parkinson's disease

 Freezes in doorways

 Lives with his daughter

**FoGO**

**Kamala, 64**

 Knee osteoarthritis

 30 km from the clinic

 Pays out of pocket

**SwaKnee**

SLIDE 5 • OPENING • 4:00–5:00 • 1.0 MIN

Two people we will follow

Visual Persona cards: Ramesh (FoGO) and Kamala (SwaKnee) with device photographs

Interaction —

Sources Composite personas (teaching device)

Each stage must answer a patient's question



We follow Ramesh and Kamala along this path.

SLIDE 6 • OPENING • 5:00–5:45 • 0.75 MIN

Each stage must answer a patient's question

Visual Six-node patient journey (editable): Need, Evidence, People, Permission, Access, Safety

Interaction —

Sources Workshop synthesis from ICMR 2017 principles and the Biodesign process

1 • NEED AND CLAIM

Claim wording sets the evidence bar



SLIDE 7 • 1 NEED AND CLAIM • 5:45–7:15 • 1.5 MIN

Claim wording sets the evidence bar

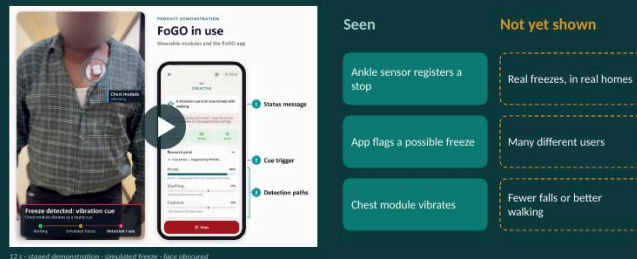
Visual Stanford need-statement formula card; two claims → consequences

Interaction Quick call-out: rewrite one claim

Sources Stanford Biodesign Need Statements brief (2022); CDSCO intended-use and software guidance; Simon, Shachar & Cohen, JAMA 2022

2 • EVIDENCE • FoGO

What did the prototype actually show?



SLIDE 8 • 2 EVIDENCE • 7:15–8:30 • 1.25 MIN

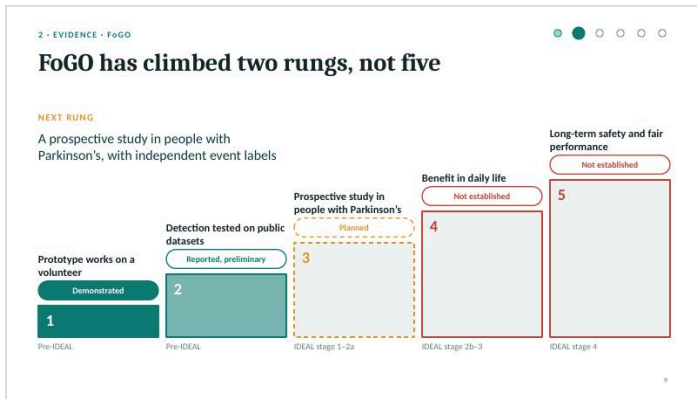
What did the FoGO prototype actually show?

Visual Embedded 12 s authentic prototype clip; Seen vs Not yet shown columns

Interaction "What did you see? What did you infer?"

Evidence status Demonstrated: staged prototype function. Not yet shown: real freezes, many users, fewer falls

Sources FoGO project deck (BIRAC BIG), supplied



SLIDE 9 • 2 EVIDENCE • 8:30–10:00 • 1.5 MIN

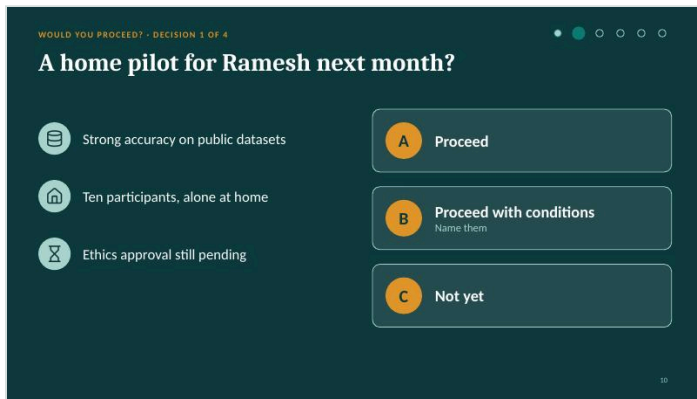
FoGO has climbed two rungs, not five

Visual Evidence ladder with status chips mapped to IDEAL stages

Interaction Which rung matters most before home use?

Evidence status 1 Demonstrated • 2 Reported, preliminary (both Pre-IDEAL) • 3 Planned (IDEAL 1–2a) • 4–5 Not established

Sources FoGO project deck (BIRAC BIG); IDEAL (McCulloch et al., Lancet 2009) and IDEAL-D (Sedrakyan et al., BMJ 2016); Cambridge Institute for Biomedical Innovation (2026) on trial-ready batches



SLIDE 10 • 2 EVIDENCE • 10:00–12:15 • 2.25 MIN

Would you proceed? A home pilot next month

Visual Decision slide: three facts + A/B/C cards

Interaction DECISION 1 (A Proceed • B With conditions • C Not yet); pairs name conditions

Sources ICMR 2017; MDR-2017 clinical investigation; MRCT post-trial access framework (2025)



SLIDE 11 • 2 EVIDENCE • 12:15–13:45 • 1.5 MIN

Rank each failure by harm and likelihood

Visual Editable 3×3 risk map with six FoGO failure modes

Interaction Which control would you test first?

Evidence status Illustrative ranking; no FoGO failure rates implied

Sources ISO 14971:2019 approach; Engineering Better Care risk perspective; Gerke et al. 2020 system view

2 • EVIDENCE • SWAKNEE

A device is also a daily routine



≈34 hours
45 minutes a day for 45 days

Who helps Kamala when something goes wrong?

1 Fit the applicator

20 s • company demonstration of use • not clinical outcome footage

12

SLIDE 12 • 2 EVIDENCE • 13:45–15:00 • 1.25 MIN

A device is also a daily routine

Visual Embedded 20 s authentic SwaKnee how-to-use clip (fit • switch on • rest); “≈34 hours” stat

Interaction Who helps Kamala when something goes wrong?

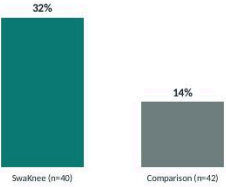
Evidence status Company demonstration of use; not clinical-outcome footage

Sources SwaKnee leaflet (45 min/day × 45 days); company video assets

2 • EVIDENCE • SWAKNEE

Which claim can this evidence carry?

Average pain reduction at day 45, company-reported (%)



Group	n	Average pain reduction (%)
SwaKnee	40	32%
Comparison	42	14%

“Less pain on average in one 45-day company study” **Reported**

“Clinically proven” **Needs independent trials**

“Regrows cartilage” **Not measured**

13

SLIDE 13 • 2 EVIDENCE • 15:00–17:00 • 2.0 MIN

Which claim can this evidence carry?

Visual Native chart 32% vs 14% (company-reported, n=40 vs 42) + claim ladder with status chips

Interaction One defends, one challenges “regrows cartilage”

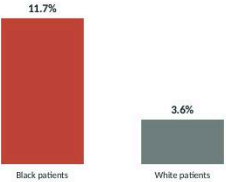
Evidence status Reported (company, not peer-reviewed) • “Clinically proven” not established • cartilage regrowth not measured

Sources Swayogya evidence page; PEMF systematic reviews; OARSI/ACR/NICE guidance

2 • EVIDENCE • EQUITY

An average can hide a patient

Oximeter read 92–96% but arterial saturation was <88%



Group	Percentage
Black patients	11.7%
White patients	3.6%

Who is missing from our data?

- Ramesh: walking aids, crowded homes
- Kamala: obesity, other illnesses, rural care
- Report results by subgroup

14

SLIDE 14 • 2 EVIDENCE • 17:00–18:15 • 1.25 MIN

An average can hide a patient

Visual Native chart 11.7% vs 3.6% occult hypoxaemia; “Who is missing?” prompts

Interaction Name one group your study could miss

Sources Sjoding et al., NEJM 2020; MRCT diversity guidance; Stanford Biodesign JEDI brief

WOULD YOU PROCEED? - DECISION 2 OF 4

Ramesh's neurologist is also the inventor

- "Will saying no change my care?"
- Consent form in English only
- Clinic wants to film him for a talk

A Recruit as planned

B Proceed with safeguards
Which ones?

C Not yet

SLIDE 15 • 3 PEOPLE • 18:15–20:30 • 2.25 MIN

Would you proceed? Ramesh's neurologist is also the inventor

Visual Decision slide with patient quote, language and filming facts

Interaction DECISION 2; role-play patient/researcher in 60 s

Sources Declaration of Helsinki 2024; ICMR 2017 consent; MRCT plain-language glossary; key-information-first consent

3 • PEOPLE • CONFLICTS OF INTEREST

Disclosure starts the work; management finishes it

Interests

- Equity or royalties
- Grants and institution
- Reputation

Risks

- Pressure to enrol
- Optimistic outcome reading
- Selective reporting

Safeguards

- Independent consent and assessment
- Prospective CTRI registration
- Publish all results

SLIDE 16 • 3 PEOPLE • 20:30–21:30 • 1.0 MIN

Disclosure starts the work; management finishes it

Visual Interests → Risks → Safeguards infographic

Interaction Which safeguard is cheapest to add today?

Sources Chao, Riskin & Krummel (Virtual Mentor 2010); Stanford GSB COI case (2011); Cambridge COI policy ('reasonably perceived'); Johnson & Rogers 2012; UCMPPMD 2024; CTRI

4 • PERMISSION • INDIA

Each permission answers a different question

Test licence
Make units for testing?
MD-12 → MD-13

Clinical investigation
Test in people?
MD-22 → MD-23 • ethics committee • CTRI

Manufacturing licence
Sell for this use?
Class A/B: State • C/D: Central

Post-market duties
Still safe?
Complaints • PSUR • MvPI • recall

Risk class A B C D Class sets the authority and the evidence expected

Compare US: IDE → 510(k) / De Novo / PMA • EU: clinical evaluation → CE mark → post-market follow-up

SLIDE 17 • 4 PERMISSION • 21:30–23:30 • 2.0 MIN

Each permission answers a different question

Visual Four-step CDSCO pathway (test licence → clinical investigation → manufacturing licence → post-market), risk-class strip, US/EU comparator

Interaction Where is Ramesh's device on this path today?

Sources MDR-2017 (Forms MD-12/13, MD-22/23, MD-3/5, MD-7/9); CDSCO Medical Device Software guidance (21 Jul 2026); US 21 CFR 812 / EU MDR comparison



SLIDE 18 • 5 ACCESS • 23:30–24:45 • 1.25 MIN

Who can't use it? Audit the demands

Visual Editable inclusive-design audit: seven capability demands (vision → dexterity) rated for FoGO and SwaKnee + 'Add for India' panel (language, literacy, power, smartphone, caregiver)

Interaction Pick the one demand you would redesign first

Evidence status Ratings illustrative, for discussion

Sources Cambridge Engineering Design Centre, Inclusive Design Toolkit; MRCT Accessibility by Design toolkit (2023)



SLIDE 19 • 5 ACCESS • 24:45–26:15 • 1.5 MIN

The price tag is only part of the cost

Visual Kamala's cost journey (buy/rent → travel → 34 h → family time → repairs) + NPPA 2017 precedent

Interaction Who pays for each step?

Sources NPPA ceiling prices (stents Feb 2017; knee implants Aug 2017); Cambridge Judge CCHLE (develop → integrate → make affordable); HBS Aravind/Aurolab case

WOULD YOU PROCEED? • DECISION 3 OF 4

Launch with this brochure?

Clinically proven!
Regrows cartilage
Doctor recommended
Avoid surgery forever

HYPOTHETICAL

Distributor wants these claims

Investors want launch this quarter

A Launch as written

B Launch, claims matched to evidence

C Not yet

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SLIDE 20 • 5 ACCESS • 26:15–28:15 • 2.0 MIN

Would you proceed? Launch with this brochure

Visual Hypothetical brochure mock-up + pressures + A/B/C

Interaction DECISION 3; rewrite one claim to match evidence

Sources UCMPMD 2024; Drugs and Magic Remedies Act 1954; ASCI health-claim guidance

WOULD YOU PROCEED? - DECISION 4 OF 4

Push tonight's FoGO update?



- Fewer false cues
- May miss more freezees
- Tested on stored data only

A Push to all users

B Staged release after review
Monitor, roll back

C Not yet

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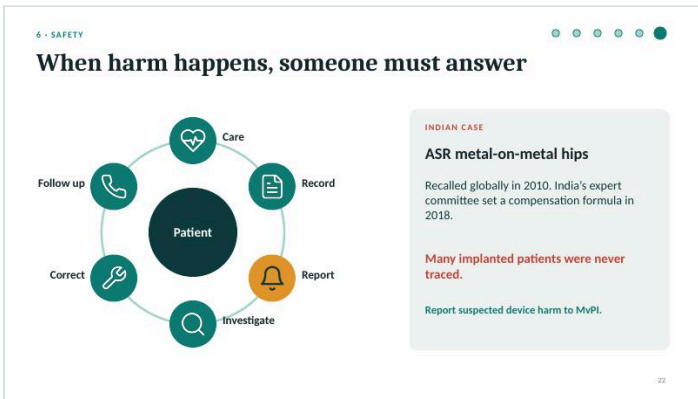
SLIDE 21 • 6 SAFETY • 28:15–29:45 • 1.5 MIN

Would you proceed? Push tonight's FoGO update

Visual Authentic app screenshot + trade-off facts + A/B/C

Interaction DECISION 4

Sources CDSCO MDSW guidance 2026 (change management, rollback); FDA PCCP guidance (Dec 2024); Gerke et al. 2020 system view; Kellmeyer et al. 2016 (closed-loop accountability)



SLIDE 22 • 6 SAFETY • 29:45–31:00 • 1.25 MIN

When harm happens, someone must answer

Visual Six-node response loop around the patient; Indian ASR hip recall case card

Interaction Assign an owner to each node

Sources MvPI (IPC); CDSCO ASR expert committee (2018); Kramer et al. 2013 post-market surveillance

CLOSING VOTE

Would you let a patient use it now?

Which patient?

Which use?

Which evidence?

Who is responsible?

Did your vote change? Why?

1 Routine care

2 Research study

3 Not yet

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SLIDE 23 • CLOSE • 31:00–32:30 • 1.5 MIN

Would you let a patient use it now? (closing vote)

Visual Four framing questions + same three answer cards as slide 3

Interaction CLOSING VOTE; compare with opening split

Sources —

Commit to one change in your own project

Before [next step], I will [action].

Owner: [name]

I proceed only if [condition].



Write it. Read it to a neighbour. Keep it.

Use the checklist on the next slide.

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SLIDE 24 • CLOSE • 32:30–34:30 • 2.0 MIN

Commit to one change in your own project

Visual Commitment card: Before __ I will __ • Owner • Proceed only if __

Interaction TAKEAWAY EXERCISE (90 s write + share with neighbour)

Sources Stanford Principled Decision-Making (team values)

Patient-impact decision checklist

Stage	Ask before proceeding	Evidence to see	Owner	Go?
Need	Is the need validated with patients and clinicians?	Need statement, observation notes	Clinical lead	☐ Yes ☐ Not yet
Evidence	Does evidence match the exact claim and users?	Protocol, subgroup results, failure modes	Study lead	☐ Yes ☐ Not yet
People	Can people refuse freely? Are conflicts managed?	Consent in local language, COI plan	Investigator, ethics committee	☐ Yes ☐ Not yet
Permission	Which CDSCO permission covers this use?	Class, licence or permission, CTRI	Regulatory lead	☐ Yes ☐ Not yet
Access	Can intended users afford, use and maintain it?	Total cost, training, service plan	Product and service	☐ Yes ☐ Not yet
Safety	Who records, reports and acts on harm?	Complaint log, MvPI route, update plan	Manufacturer, clinic	☐ Yes ☐ Not yet

For every "Not yet": name the owner and the condition for proceeding.

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SLIDE 25 • CLOSE • 34:30–35:00 • 0.5 MIN

Patient-impact decision checklist

Visual Editable 6-row table: stage • ask • evidence to see • owner • go/not yet

Interaction Reusable tool; printed handout

Sources Workshop synthesis

Evidence to request before stronger claims

Question	FoGO	SwaKnee
Outcome	Freeze detection against video-annotated events; falls, confidence, walking	Pain and function with minimal important difference; structural outcomes only if claimed
Comparison	Participant-level validation; cue vs no-cue or sham cue	Randomised, blinded sham comparison; co-interventions recorded
Who	Disease stage, walking aids, homes, turning and doorways	Age, BMI, severity, comorbidity, rural access
Burden and harm	False cues, missed events, signal loss, skin, data	Adverse events, adherence to 45-minute sessions, support needs
Version	Firmware and model version for every result	Controller and applicator version for every result

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SLIDE 26 • APPENDIX • REFERENCE

Evidence to request before stronger claims (FoGO vs SwaKnee)

Visual Reference table

Interaction Back-pocket for Q&A

Sources As cited

Sources (draft)

[Example](#)
detail

Sources: teaching materials, Indian regulation, evidence, media credits

Visual Clickable reference slides

Interaction —

Sources Full list